

ST MICHAEL LOEBEN, Austria

WMO: 111740

Lat: 47.333N

Lon: 15.000E

Elev: 572

StdP: 94.64

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.5	-15.4	1.0	-12.5	-13.3	1.3	-9.9	7.1	4.7	6.2	4.4	0.6	250	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	29.8	19.7	27.8	19.0	26.0	18.1	20.7	27.6	19.7	26.1	18.9	24.4	2.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	14.3	22.7	17.6	13.5	22.0	16.8	12.9	21.1	62.2	27.7	58.8	26.2	55.9	24.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.0	4.1	-16.7	33.0	3.0	1.7	-18.9	34.2	-20.6	35.2	-22.2	36.2	-24.4	37.4		
(5)			-17.0	22.5	2.8	1.2	-19.1	23.4	-20.7	24.1	-22.3	24.7	-24.3	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-2.5	0.4	4.4	8.9	13.7	16.9	18.7	18.3	14.1	9.2	3.8	-1.4	(6)
(7)		DBStd	8.24	4.14	4.05	4.03	3.72	3.32	3.50	3.11	2.89	3.29	3.82	3.99	3.85	(7)
(8)		HDD10.0	1509	388	269	176	63	6	1	0	0	5	61	189	353	(8)
(9)		HDD18.3	3608	646	503	432	283	146	67	34	36	132	282	437	611	(9)
(10)		CDD10.0	1054	0	0	2	30	122	209	269	257	127	37	2	0	(10)
(11)		CDD18.3	111	0	0	0	0	4	24	44	34	4	0	0	0	(11)
(12)		CDH23.3	1101	0	0	0	6	82	262	404	310	37	1	0	0	(12)
(13)		CDH26.7	283	0	0	0	0	13	66	119	83	3	0	0	0	(13)
(14)	Wind	WSAvg	1.3	1.3	1.5	1.6	1.6	1.5	1.3	1.4	1.1	1.0	1.1	1.3	1.2	(14)
(15)	Precipitation	PrecAvg	1020	45	45	58	63	102	135	148	134	99	78	64	49	(15)
(16)		PrecMax	1250	107	93	156	127	177	197	322	222	150	157	119	96	(16)
(17)		PrecMin	767	9	8	18	11	53	76	44	63	51	18	0	5	(17)
(18)		PrecStd	115	27	24	28	25	34	34	55	43	27	34	26	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	13.7	18.6	23.8	28.4	31.0	32.3	32.0	26.8	21.9	17.6	9.1	(19)
(20)			MCWB	5.7	7.7	10.7	14.4	17.8	20.2	20.7	20.2	18.3	15.5	12.1	5.7	(20)
(21)		2%	DB	6.5	10.6	15.9	21.1	25.5	28.7	29.8	29.0	24.1	19.0	12.9	6.1	(21)
(22)			MCWB	4.1	5.8	9.3	12.7	16.6	19.1	20.2	19.3	17.0	14.0	9.9	4.3	(22)
(23)		5%	DB	4.4	8.1	13.5	18.8	23.5	26.6	27.7	26.7	22.0	16.7	10.9	4.3	(23)
(24)			MCWB	2.4	4.6	8.2	11.7	15.5	18.3	19.3	18.6	15.9	12.7	8.5	3.0	(24)
(25)		10%	DB	2.7	6.1	11.1	16.4	21.2	24.4	25.5	24.7	20.0	14.8	9.1	3.1	(25)
(26)			MCWB	1.3	3.3	7.1	10.3	14.4	17.3	18.3	17.9	15.1	11.8	7.3	2.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	8.1	11.7	15.0	18.6	21.4	22.1	21.4	19.0	16.3	12.1	6.6	(27)
(28)			MCD B	9.0	13.4	16.9	22.8	26.0	28.2	29.2	28.3	24.9	20.3	16.3	8.4	(28)
(29)		2%	WB	4.2	6.2	9.7	13.5	17.2	20.1	20.8	20.2	17.8	14.8	10.3	4.5	(29)
(30)			MCD B	6.3	10.0	14.9	19.5	23.9	26.8	28.1	27.0	22.7	18.2	12.7	5.7	(30)
(31)		5%	WB	2.6	4.9	8.5	12.2	16.1	19.0	19.8	19.2	16.7	13.5	9.0	3.2	(31)
(32)			MCD B	4.1	7.7	13.0	17.5	22.2	25.0	26.2	25.2	20.6	16.1	10.6	4.1	(32)
(33)		10%	WB	1.5	3.5	7.3	10.9	15.0	17.9	18.9	18.3	15.6	12.1	7.6	2.2	(33)
(34)			MCD B	2.4	5.8	10.6	15.3	20.1	23.0	24.2	23.5	19.0	14.2	8.8	3.0	(34)
(35)	Mean Daily Temperature Range	MDBR		6.9	9.1	10.1	11.1	11.6	11.1	11.3	11.1	10.4	9.2	6.5	5.7	(35)
(36)		5% DB	MCD BR	8.1	12.4	15.1	16.2	16.2	15.7	15.3	15.5	13.5	11.8	8.9	6.6	(36)
(37)			MCWBR	5.6	7.9	8.8	8.3	7.6	6.8	6.3	6.5	6.7	7.0	5.9	4.9	(37)
(38)		5% WB	MCD BR	7.4	11.5	14.2	14.4	14.9	14.0	14.0	13.8	12.3	10.8	8.0	5.9	(38)
(39)			MCWBR	5.3	7.6	8.5	7.6	7.2	6.4	6.1	6.1	6.3	6.7	5.5	4.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.290	0.319	0.362	0.407	0.414	0.420	0.422	0.419	0.387	0.357	0.325	0.291	(40)
(41)		taud		2.404	2.342	2.289	2.226	2.258	2.279	2.275	2.301	2.362	2.422	2.463	2.462	(41)
(42)		Ebn at Noon		798	842	853	844	852	846	838	825	818	789	749	756	(42)
(43)		Edh at Noon		68	90	111	130	131	129	128	120	103	83	66	58	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.21	2.07	3.12	4.13	4.83	5.30	5.29	4.72	3.41	2.33	1.30	1.00	(44)
(45)		RadStd		0.13	0.26	0.36	0.44	0.53	0.40	0.46	0.54	0.44	0.23	0.19	0.14	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (109 neighbors)		+0.55	N/A	N/A	+0.89	+0.47	+0.40	-108	-145	+92	+55	

Nomenclature: See separate page